The Solar Cycle – Background Information and Additional Resources

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Background information:

- The Earth has a **magnetic field** that is mostly generated by the movement of liquid iron in the Earth's outer core in a process known as the geodynamo (NASA, 2021).

- The magnetic field extends out from Earth into space and forms the **magnetosphere.** The magnetosphere acts as a huge protective 'force field' and shields the planet from harmful particles emitted by the Sun (Olsen & Pauluhn, 2019) (See **Figure 1**).

- The aurora is a spectacular naturally occurring light display within Earth's upper atmosphere (BoM SWS). ***Aurora australis*** refers to the aurora in the southern hemisphere (the southern lights); ***Aurora borealis*** refers to the aurora in the northern hemisphere (the northern lights) (NASA, 2025).

- Auroras occur when the Sun releases high energy charged particles in events such as coronal mass ejections (CMEs) (BoM SWS) (see **Figure 1**). These solar particles interact with Earth's magnetosphere, causing **geomagnetic activity** and the emission of light, which is seen as the aurora.

- Geoscience Australia has 10 permanent **geomagnetic observatories** – 6 on the mainland (including Canberra), 2 in Antarctica, 1 on Macquarie Island and 1 in the Cocos Islands (Geoscience Australia, 2025). These observatories each contain multiple **magnetometers** that monitor Earth's geomagnetic activity in real-time.

- Geomagnetic activity is quantified using geomagnetic indices. Two important indices are the **k index** and the **aa index** (Geoscience Australia).

- The **k index** measures geomagnetic activity every 3 hours relative to a quiet day on a scale of 0-9 for a specific observatory (Geoscience Australia).

- The **aa index** is an average measure of geomagnetic activity that uses the k index values from two observatories on opposite sides of the world – Canberra, Australia and Hartland, England (Geoscience Australia).

- **Sunspots** are dark, cooler regions on the surface of the Sun caused by intense magnetic activity (NASA, 2025). They are the source of solar eruptions like CMEs, which drive geomagnetic activity on Earth. Sunspot number varies over time in a natural cycle.

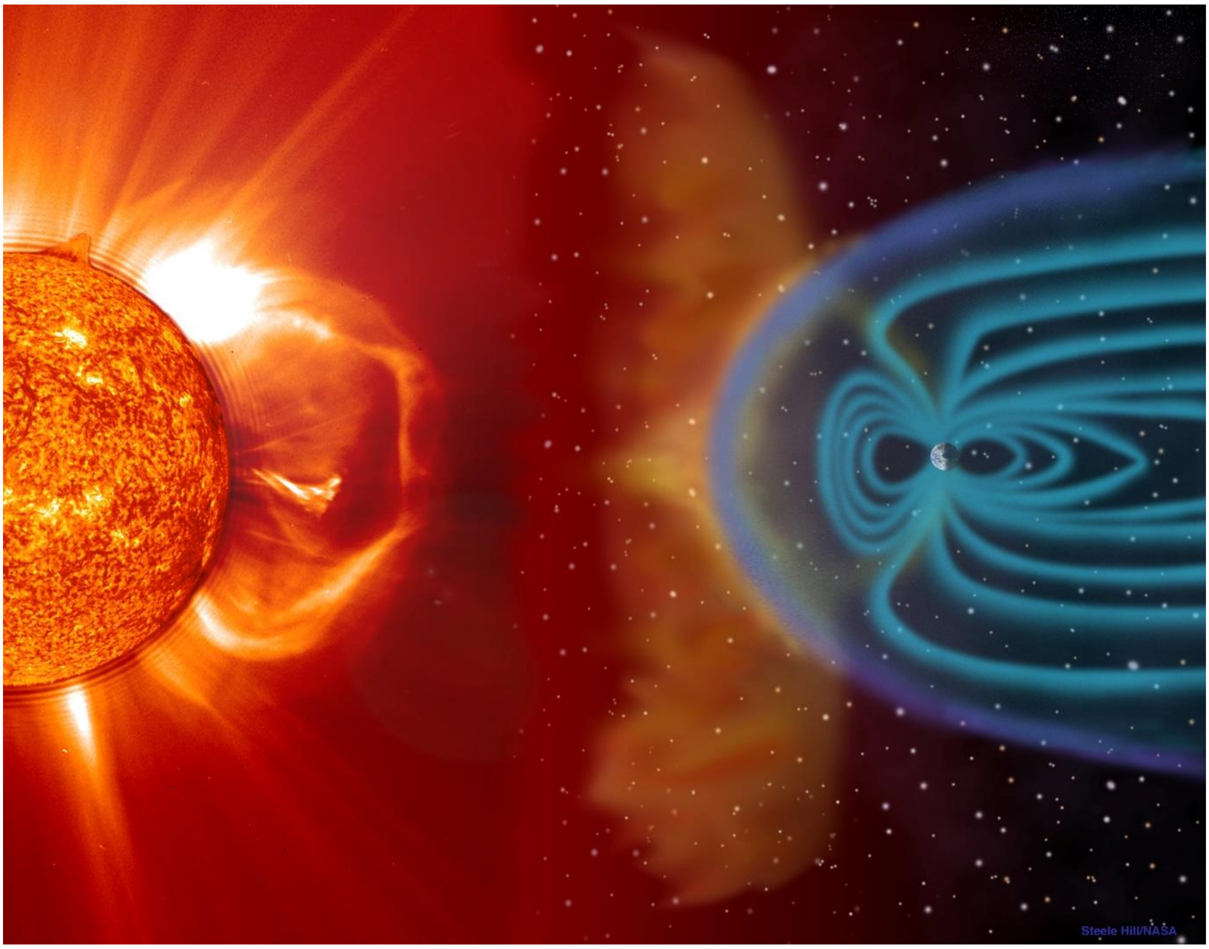

Figure 1: The Earth's magnetosphere acts as a protective shield around the planet and shields it from harmful particles emitted from the Sun in events such as CMEs (European Space Agency - ESA).

References:

Bureau of Meteorology Australian Space Weather Forecasting Centre (BoM SWS). *Aurora Photos and the Aurora Explained*. Available at: https://www.sws.bom.gov.au/Educational/1/1/4

Geoscience Australia. *Geomagnetic Indices: aa index*. Available at: https://geomagnetism.ga.gov.au/geomagnetic-indices/aa-index

Geoscience Australia. *Geomagnetic Indices: k index*. Available at: https://geomagnetism.ga.gov.au/geomagnetic-indices/k-index

Geoscience Australia (2025). *Space weather.* Available at: https://www.ga.gov.au/education/natural-hazards/space-weather

NASA (2025). *Auroras*. Available at: https://science.nasa.gov/sun/auroras/

NASA (2021). *Earth's Magnetosphere: Protecting Our Planet from Harmful Space Energy.* Available at: https://science.nasa.gov/science-research/earth-science/earths-magnetosphere-protecting-our-planet-from-harmful-space-energy/

NASA (2025). *Sunspots*. Available at: https://science.nasa.gov/sun/sunspots/

Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at: https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

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Additional resources:

Bureau of Meteorology Aurora webpage, including current aurora conditions and two great introductory videos 'what is the aurora' and 'catching the aurora': Bureau of Meteorology Australian Space Weather Forecasting Centre (SWS BoM). *Aurora*. Available at: https://www.sws.bom.gov.au/Aurora

For more information about Geoscience Australia's network of geomagnetic observatories: Geoscience Australia (2025). *Space weather.* Available at: https://www.ga.gov.au/education/natural-hazards/space-weather

NOTE: The following resource contains answers relating to the class. Direct students to this AFTER they have completed the class:

NASA Sunspots webpages, including sunspot animations, videos and sunspot news: NASA. *Sunspots*. Available at: https://science.nasa.gov/sun/sunspots/